Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 2

Part A

i) mol H_2O = 2.883g H_2O × $\frac{1 \text{ mol } H_2O}{18.02 \text{ g } H_2O}$

= 0.1600 mol H_2O

ii) mol H = 0.1600 mol H_2O × $\frac{2 \text{ mol H}}{1 \text{ mol } H_2O}$ = $\frac{0.32 \text{ mol}}{0.24 \text{ mol}}$ = 1.3 × 3 = 4

mol C = 0.2400 mol CO_2 × $\frac{1 \text{ mol C}}{1 \text{ mol } CO_2}$ = $\frac{0.24 \text{ mol}}{0.24 \text{ mol}}$ = 1 × 3 = 3

∴ $C_3H_4O_3$

Part B

i) pOH = -log(0.0550 M)

= 1.26

pH = 12.74

[HAsc] = 1.82×10^{-13} M

ii) pKa = 7.5

iii) $\frac{1}{3}$

Question 2 is continued on the next page.

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 2

Part C

i) The concentration from trial 1 to 3 for [HAsc] doubles while the initial rate of formation for those same trials also doubles. The represents a first reaction order because both the concentration and the initial rate doubled instead of quadrupling or staying the same.

ii) Rate = k [HAsc][I2]

2.457×10^{-4} M/s = k [0.450M][0.200M]

2.457×10^{-4} M/s = 0.54 M^2 k

$\boxed{k = 4.55 \times 10^{-4}\ S/M}$

Part D

I3 and water contain both london dispersion forces as well as dipole-dipole forces. on the other hand, I2 and water only contain london dispersion forces.

Continue to Question 3.